

ROBOTICS AND AI

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Class X

Assignment 8–12

Due Date: 07.10.26

50 marks

Assignment 8 Sensors in Robotics

Question 1: List and explain any four types of sensors used in robotics with one example each. (3 marks)

Question 2: Differentiate between Internal and External sensors in a robot. Give two examples of each type. (3 marks)

Question 3: Describe how a proximity sensor and a vision sensor function in a warehouse robot to ensure safe navigation. (4 marks)

Assignment 9 Actuators

Question 1: Define actuators and state their importance in robotic systems. (3 marks)

Question 2: Classify actuators into linear and rotary types. Explain each with suitable examples. (3 marks)

Question 3: Explain how actuators work in combination with sensors to perform a pick-and-place operation in an industrial robot. (4 marks)

Assignment 10 Controller in Robotics

Question 1: What is a controller in a robotic system? State its primary function. (3 marks)

Question 2: Differentiate between manual and automatic control systems with examples. (3 marks)

Question 3: Draw and label a block diagram showing Input → Controller → Robot → Feedback, and explain how the system maintains accuracy. (4 marks)

Assignment 11

Integration of Sensors, Actuators and Controller

Question 1: Explain how sensors, actuators, and controllers are interrelated in a robotic system. (3 marks)

Question 2: Using an example, describe how the angular position of a robotic arm is measured and controlled. (3 marks)

Question 3: Design a simple conceptual workflow showing how input data from sensors is processed by the controller to drive actuators in a mobile robot avoiding obstacles. (4 marks)

Assignment 12

Python: Control Flow

Question 1: Write a Python program to input three numbers and find the greatest and smallest among them. *Hint: Use conditional statements (if, elif, else). Don't use max(), min()* (3 marks)

Question 2: Write a Python program that inputs marks of five subjects, calculates the average, and displays the grade according to the following rules: (4 marks)

Average Marks	Grade
90 – 100	A+
80 – 89	A
70 – 79	B
60 – 69	C
50 – 59	D
Below 50	Fail

Question 3: Write a Python program to print a pyramid pattern of stars (*) using loops. *Hint: Use nested for loops — one for rows and another for spaces and stars.* (3 marks)

Example: If number of rows = 5, then output:

```
  *
 ***
*****
*****
*****
```